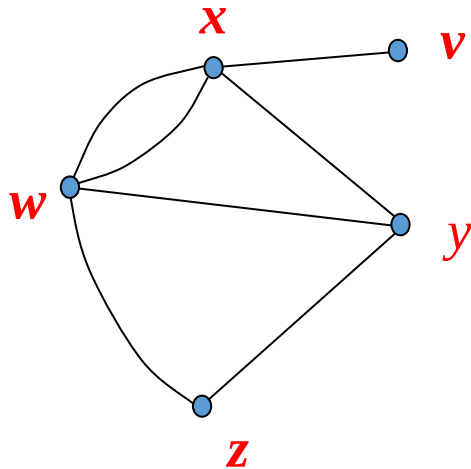


Degree Sequence

- The *Degree Sequence* of a graph is the list of vertex degrees, usually written in non-increasing order, as $d_1 \geq \dots \geq d_n$. (**WARNING:** M&N uses the term **score** for the degree sequence, and lists the degrees in non-decreasing order.) As we can see from the example below, the degree sequence (score) is meaningful for general graphs, but we will only consider simple graphs.



Example:

Degree sequence:

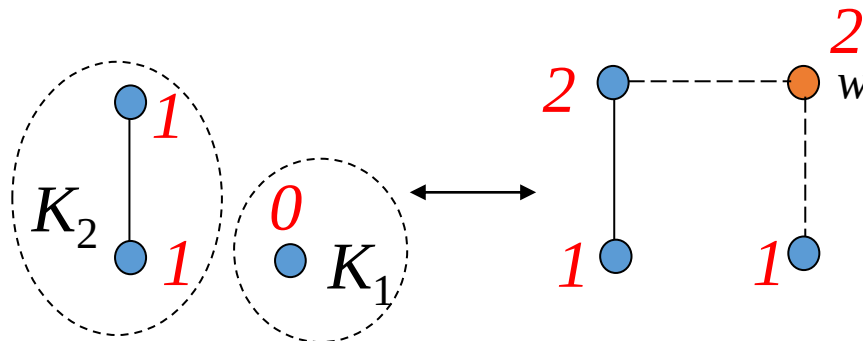
$d(w), d(x), d(y), d(z), d(v)$
or 4,4,3,2,1

Graphic Sequence

- A *graphic sequence* is a list of nonnegative numbers that is the degree sequence (score) of some *simple* graph.
- A simple graph “realizes” d .
 - means: A simple graph with degree sequence d .

Recursive condition

- The lists $(2, 2, 1, 1)$ and $(1, 0, 1)$ are graphic. The graph $K_2 + K_1$ realizes $1, 0, 1$.
- Adding a new vertex adjacent to vertices of degrees 1 and 0 yields a graph with degree sequence $2, 2, 1, 1$, as shown below.
- Conversely, if a graph realizing $2, 2, 1, 1$ has a vertex w with neighbors of degrees 2 and 1, then deleting w yields a graph with degrees $1, 0, 1$.



Recursive condition

- Similarly, to test 33333221, we seek a realization with a vertex w of degree 3 having three neighbors of degree 3.

Delete this Vertex

3 3 3 3 3 2 2 1

A new degree sequence

2 2 2 3 2 2 1

Recursive condition

This exists if and only if 2223221 is graphic. (See next page)

- We reorder this and test 3222221.
- We continue deleting and reordering until we can tell whether the remaining list is realizable.
- If it is, then we insert vertices with the desired neighbors to walk back to a realization of the original list.
- The realization is **not** unique. (**EXERCISE !**)
- Nevertheless, the next theorem implies that this recursive test works to determine whether there is a simple graph (**in other words, at least one simple graph**) realizing a given non-decreasing sequence (of non-negative integers).

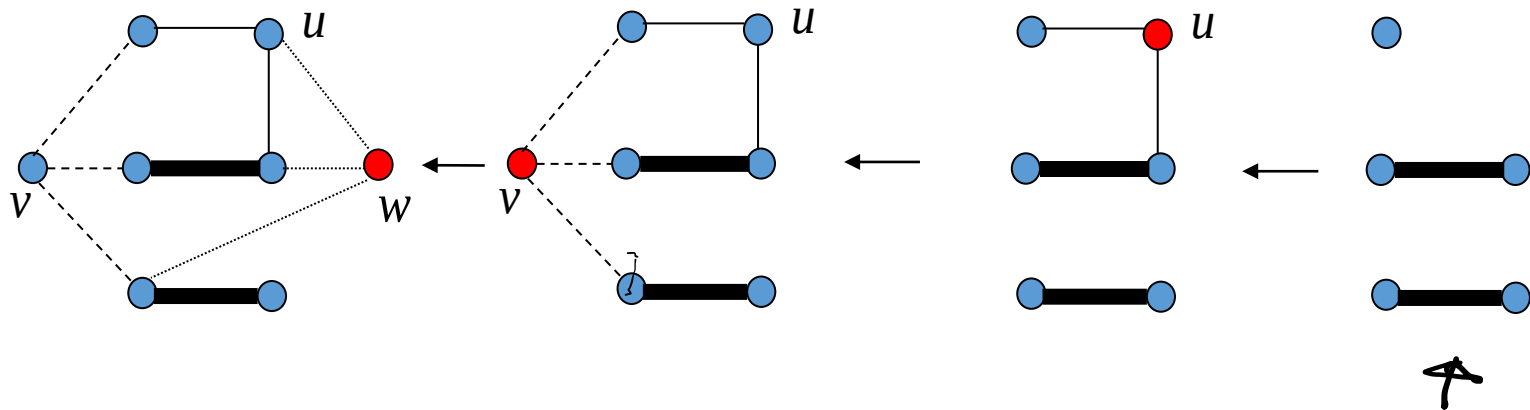
Recursive condition

3333221
2223221

3222221
111221

221111
10111

11110



Necessary and Sufficient Condition for a Sequence to be the Degree Sequence (Score) of a Simple Graph

- **Theorem 7 (Havel-Hakimi Theorem):** For $n > 1$, an integer list d of size n is graphic if and only if d' is graphic, where d' is obtained from d by deleting its largest element Δ and subtracting 1 from its Δ next largest elements. The only 1-element graphic sequence is $d_1 = 0$.
- **Remark:** The statement in M&N is equivalent to the above, but this version is slightly easier to understand and apply.

Theorem 7: For $n > 1$, an integer list d of size n is graphic if and only if d' is graphic, where d' is obtained from d by deleting its largest element Δ and subtracting 1 from its Δ next largest elements. The only 1-element graphic sequence is $d_1 = 0$.

Proof: 1/6

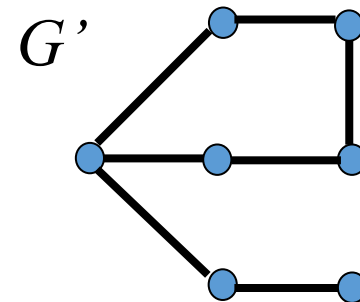
- For $n = 1$, the statement is trivial.
- For $n > 1$, **we first prove that the condition is sufficient.**
 - Given d with $d_1 \geq \dots \geq d_n$ and a simple graph G' with degree sequence d'

For Example:

We have: 1) $d = 3333221$

2) G' with $d' = 2223221$

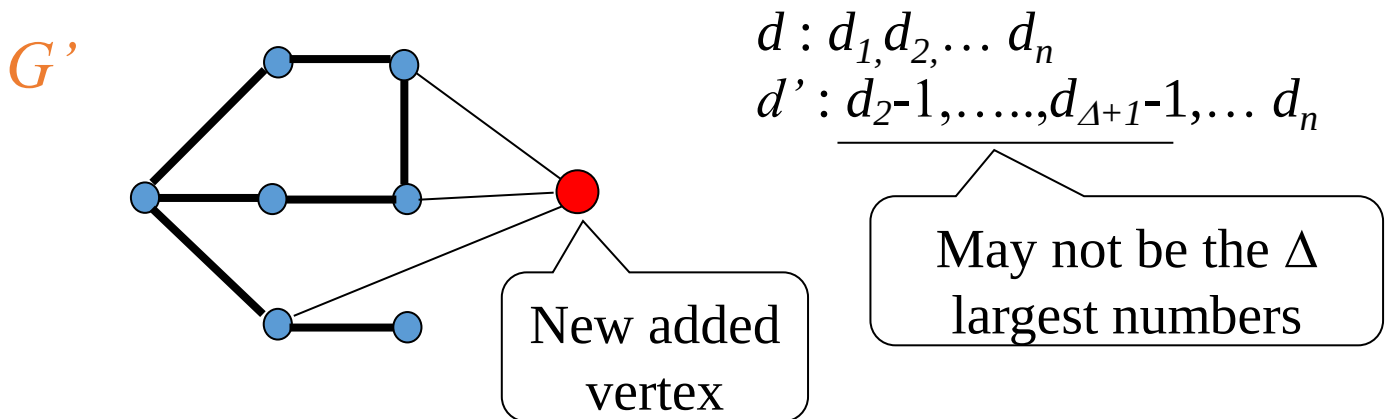
We show: d is graphic



Theorem 7: For $n > 1$, an integer list d of size n is graphic if and only if d' is graphic, where d' is obtained from d by deleting its largest element Δ and subtracting 1 from its Δ next largest elements. The only 1-element graphic sequence is $d_1 = 0$.

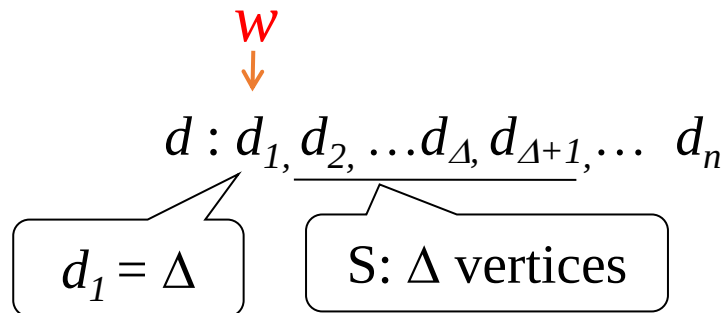
Proof: 2/6

- We add a new vertex adjacent to vertices in G' with degrees $d_2-1, \dots, d_{\Delta+1}-1$.
- These d_i are the Δ largest elements of d after (one copy of) Δ itself,
- Note : $d_2-1, \dots, d_{\Delta+1}-1$ need not be the Δ largest numbers in d' (see example in previous page)



Theorem 7: For $n > 1$, an integer list d of size n is graphic if and only if d' is graphic, where d' is obtained from d by deleting its largest element Δ and subtracting 1 from its Δ next largest elements. The only 1-element graphic sequence is $d_1 = 0$.

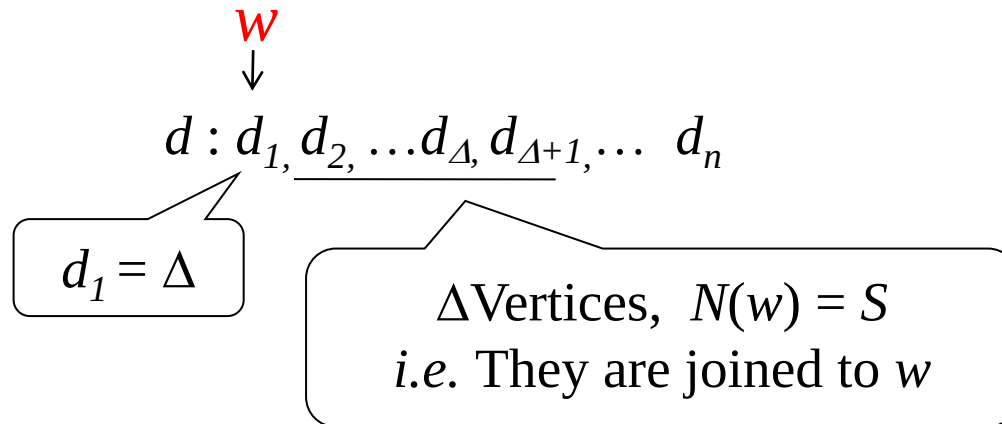
- To prove necessity, 3/6
 - Given a simple graph G realizing d , we produce a simple graph G' realizing d'
 - Let w be a vertex of degree Δ in G , and let S be a set of Δ vertices in G having the “desired degrees” $d_2, \dots, d_{\Delta+1}$



Theorem 7: For $n > 1$, an integer list d of size n is graphic if and only if d' is graphic, where d' is obtained from d by deleting its largest element Δ and subtracting 1 from its Δ next largest elements. The only 1-element graphic sequence is $d_1 = 0$.

Proof: *continue* 4/6

- Case 1: If $N(w) = S$. Then we delete w to obtain G' .



Delete w than we have

$$d' : d_2-1, \dots, d_{\Delta+1}-1, \dots, d_n$$

Theorem 7: For $n > 1$, an integer list d of size n is graphic if and only if d' is graphic, where d' is obtained from d by deleting its largest element Δ and subtracting 1 from its Δ next largest elements. The only 1-element graphic sequence is $d_1 = 0$.

There is a graph G with degree sequence d s.t. $d(w) = \Delta$

$N(w) = \text{nbhd of } w$

Proof: **continue** 5/6

- Case 2: Otherwise, $N(w) \neq S$ (Recall that $|N(w)| = \Delta = |S|$.)
- Some vertex of S is missing from $N(w)$.
 - In this case, we modify G to increase $|N(w) \cap S|$ without changing any vertex degree.
 - Since $|N(w) \cap S|$ can increase at most Δ times, repeating this converts G into another graph G^* that realizes d and has S as the neighborhood of w .
 - Now G^* satisfies Case 1. So, from G^* we then delete w to obtain the desired graph G' realizing d' .

with Δ highest degrees after w .

w



$d : d_1, d_2, \dots, d_\Delta, d_{\Delta+1}, \dots, d_n$

$d_1 = \Delta$

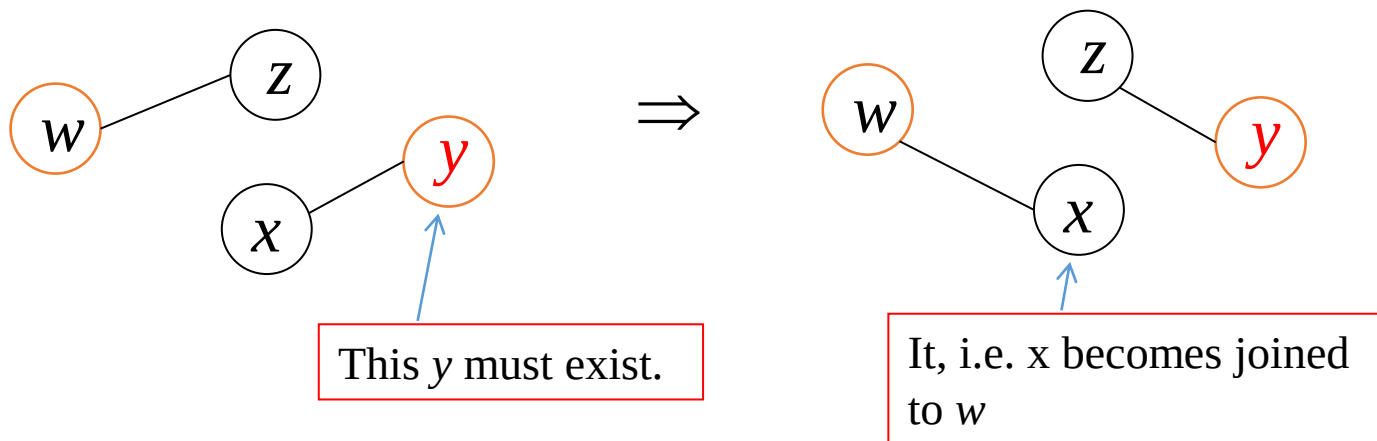
Δ Vertices, $N(w) \neq S$

i.e. Some vertices are not joined to w .
- We make them become joined to w without changing their degree.

Theorem 7: For $n > 1$, an integer list d of size n is graphic if and only if d' is graphic, where d' is obtained from d by deleting its largest element Δ and subtracting 1 from its Δ next largest elements. The only 1-element graphic sequence is $d_1 = 0$.

Proof: *continue* 6/6

- To find the modification when $N(w) \neq S$, we choose $x \in S$ and $z \notin S$ so that wz are joined and wx are not.
- We want to add edge wx and delete edge wz , but we must preserve vertex degrees. Since $d(x) \geq d(z)$ and already w is a neighbor of z but not of x , there must be a vertex y adjacent to x but not to z . Now we delete $\{wz, xy\}$ and add $\{wx, yz\}$ to increase $|N(w) \cap S|$.



To quickly go over the logic of the proof: We know that d is a graphic sequence and G realizes it.

If $d' = \text{degree sequence of } G - w,$

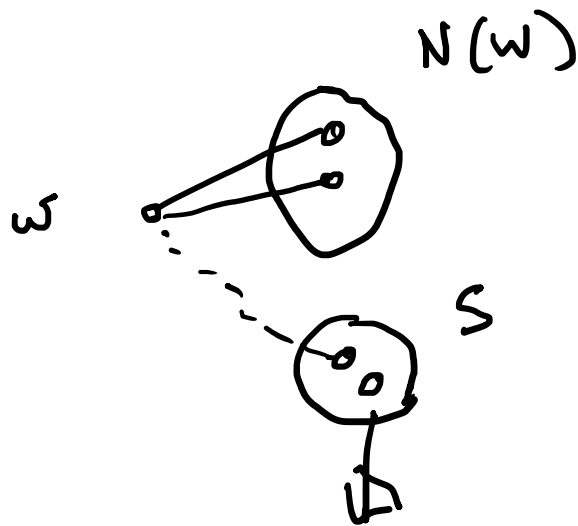
then obviously d' is graphic.

However, d' may not be the degree sequence of $G - w$. d' is obtained by reducing the degrees of vertices of highest degree, i.e. the vertices in the set S .

However, degree sequence of $G - w$ is obtained by reducing degree of vertices in $N(w)$.

If $N(w) \neq S$ then we have
 a difficulty ! ! (See diag. on next slide)
~~How~~ But we know that $|N(w)| = |S| = \Delta$.
 So, we change the graph G to another
 graph, say G_1 , on same set of vertices,
 s.t. for each $v \in V(G) = V(G_1)$, $d_G(v) =$
 $d_{G_1}(v)$, i.e. degree sequence of $G_1 =$
 degree sequence of $G = d$. Moreover, $N_G(w)$
 has one more vertex of S . Repeating
 this process at most Δ times, we get
 S in Case 1, and result follows.

Diagram :-



$x \in S$

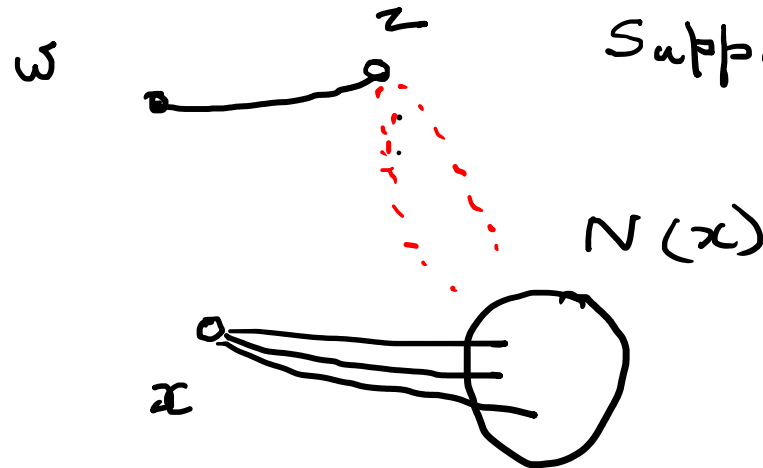
$x \notin N(w)$

~~note~~

Note: Since we don't change vertex degrees, S remains the same. S = set of vertices of highest degree after w .

However, we change $N(w)$ at each step, throwing out a "bad" vertex (low degree) and bringing in a good vertex (high degree).

Recall : $d(w) = \Delta = \max$ possible



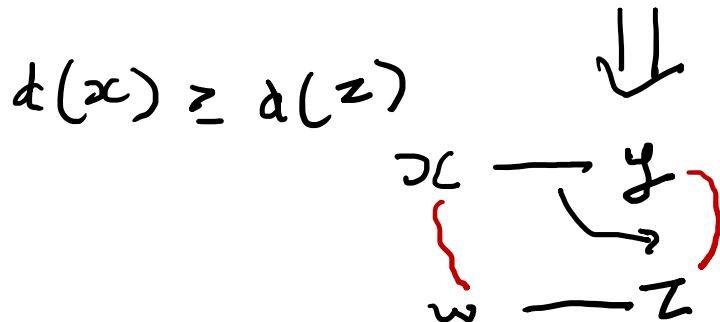
Suppose z is joined to every vertex in $N(x)$

Then : $N(z) \supseteq$

$N(x) \cup \{w\}$

$\Rightarrow d(z) > d(x)$

$\Rightarrow \Leftarrow$



$d(x) \geq d(z)$

$\rightarrow$ There was no change of degree

$G \rightarrow G_1$

$d = d_1$

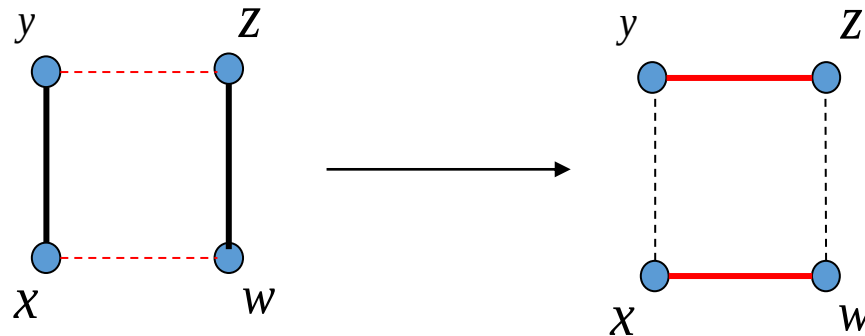
Graphs with the Same Degree Sequence

Remark: What can we say about graphs with the same degree sequence ? As we have seen by an earlier example, they are not necessarily isomorphic. However, if we consider the family $\mathbb{G}$ of all simple graphs on the same vertex set V , then having the same degree sequence is an equivalence relation on $\mathbb{G}$ (**Prove as an exercise !**). The nature of this relation is captured by the key idea in the proof of Theorem 7. We state it as a Corollary.

Corollary 7.1: If G and H are two simple graphs with vertex set V , then $d_G(v)=d_H(v)$ for every $v \in V$ if and only if there is a sequence of 2-switches that transforms G into H .

2-switch

- A **2-switch** is the replacement of a pair of edges xy and zw in a simple graph by the edges yz and wx , given that yz and wx did not appear in the graph originally.



Corollary 7.1: If G and H are two simple graphs with vertex set V , then $d_G(v) = d_H(v)$ for every $v \in V$ if and only if there is a sequence of 2-switches that transforms G into H .

Proof: (1/3)

- Every 2-switch preserves vertex degrees, so the condition is sufficient.
- Conversely, when $d_G(v) = d_H(v)$ for all $v \in V$, we obtain an appropriate sequence of 2-switches by induction on the number of vertices, n .
- If $n < 3$, then for each choice of $d_1, \dots, d_n$ there is at most one simple graph with $d(v_i) = d_i$.
- Hence we can use $n = 2$ as the basis step.

Corollary 7.1: If G and H are two simple graphs with vertex set V , then $d_G(v) = d_H(v)$ for every $v \in V$ if and only if there is a sequence of 2-switches that transforms G into H .

Proof (2/3)

- Consider $n \geq 3$, and let w be a vertex of maximum degree, Δ
- Let $S = \{v_1, \dots, v_\Delta\}$ be a fixed set of vertices with the Δ highest degrees other than w
- Some sequence of 2-switches transforms G to a graph G^* such that $N_{G^*}(w) = S$, and some such sequence transforms H to a graph H^* such that $N_{H^*}(w) = S$ (*same idea as in the proof of Theorem 7*)
- Since $N_{G^*}(w) = N_{H^*}(w)$, deleting w leaves simple graphs $G' = G^* - w$ and $H' = H^* - w$ with $d_{G'}(v) = d_{H'}(v)$ for every vertex v

Corollary 7.1: If G and H are two simple graphs with vertex set V , then $d_G(v) = d_H(v)$ for every $v \in V$ if and only if there is a sequence of 2-switches that transforms G into H .

Proof (3/3):

- By the induction hypothesis, some sequence of 2-switches transforms $\mathbf{G'}$ to $\mathbf{H'}$. Since these do not involve $\mathbf{w}$, and $\mathbf{w}$ has the same neighbors in $\mathbf{G^*}$ and $\mathbf{H^*}$, applying this sequence transforms $\mathbf{G^*}$ to $\mathbf{H^*}$.
- Hence we can transform $\mathbf{G}$ to $\mathbf{H}$ by transforming $\mathbf{G}$ to $\mathbf{G^*}$, then $\mathbf{G^*}$ to $\mathbf{H^*}$, then (in reverse order) the transformation of $\mathbf{H}$ to $\mathbf{H^*}$.